Project Proposal and Progress Statement — MAC F244 (Semester I, AY 2026–27)

Prepared 13 November 2026

> **Date note (please read).** The course document asks for the proposal and progress statement by **31 October 2026** and the final submission by **15 November**. This document was prepared on 13 November 2026 and is **not backdated**. If your group already submitted a proposal, keep that version; this one can serve as a record or be discarded. Instructor approval of the topic is **not** claimed here.

Group members

#	Name	ID number
1	[Name]	[ID]
2	[Name]	[ID]
3	[Name]	[ID]
4	[Name]	[ID]
5	[Name]	[ID]

Title

Delta Hedging of European Options: How Rebalancing Frequency, Volatility Misspecification, and Transaction Costs Affect Hedging Errors (suggested topic 12 in the course list).

Problem

An option seller receives a premium and must pay the option's payoff at maturity. Delta hedging (holding $\partial C/\partial S$ shares, adjusted over time) removes this risk perfectly only under the Black–Scholes ideal: continuous trading, no costs, known volatility. We measure the hedging error that remains when (i) the seller rebalances only N times, (ii) uses the wrong volatility, (iii) pays proportional transaction costs. We also measure (iv) how the answers change across asset volatility regimes, moneyness and maturity.

Mathematical model

- Stock: geometric Brownian motion $dS = \mu S\,dt + \sigma S\,dW$ under the physical measure P (simulation of hedging P&L); drift r under the risk-neutral measure Q (pricing).

- Option: European call, Black–Scholes price and delta from Itô's lemma, replication and the BS PDE.
- Hedge: discrete self-financing portfolio of stock and a cash account earning r; proportional cost c per side on every trade, including initial purchase and final liquidation.
- Analytic benchmarks: Kamal–Derman approximation sd ≈ √(π/4)·vega·σ/√N; gamma-weighted variance-gap formula for misspecified volatility; Leland's cost-adjusted volatility.

Computational methodology

Python (NumPy, SciPy, pandas, Matplotlib). Exact GBM simulation on a fine grid. Every rebalancing frequency is applied to the same simulated paths (common random numbers). Monte Carlo standard errors are reported for every statistic. Validation covers put–call parity, finite-difference delta, Q-Monte-Carlo price, a hand-checked ledger, boundary cases and timestep refinement. A small real-data part estimates volatility from historical prices and runs one historical-path backtest.

Expected outcomes

- Error spread ∝ N^(−1/2) with mean ≈ 0 when σ is correct.
- A bias proportional to ($\sigma_h^2 - \sigma^2$) that does not disappear with more frequent trading.
- An interior optimal rebalancing frequency once costs are included.
- Larger relative errors for out-of-the-money, short-dated, low-volatility options.

Progress status (as of 13 November 2026)

Completed (verified by running the code): core library; 22 validation checks (all pass); experiments A–E in quick and full modes; figures and CSV tables; executed notebook; 4-page report and slides (drafts for the group to review). **Remaining for the group:** insert names/IDs; check every derivation and number; adapt the text to your own words; check AI-use policy; rehearse the viva (≈15 min talk + ≈5 min questions).